NUI Galway
OÉ Gaillimh

MSc Thesis for Health Data Science

A Bayesian Time-Series Framework for Comparing Hospital Pressure Across Irish HSE Health Regions

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Declaration

We hereby certify that this material, which we now submit for assessment on the programme of study leading to the award of Master of Science is entirely our own work and had not been taken from the work of others or Artificial Intelligence, save and to the extent that such work has been cited and acknowledged within the text of our work.

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1 Introduction

2 Data

2.1 Data Collection

Figure 1: Data pipeline from source to model input.

3 Methods

3.1 Model Development

The models were developed iteratively with increasing complexity as described in Gelman (2004). Each addition to the model is motivated by a combination of a priori knowledge, residual patterns and minimising the DIC. The iterative process is illistrated in figure 2 of the appendix. The DIC comparison table, see table 1, reads as a record of model selection.

All models share the same Normal likelihood:

$$y_{i,t} \sim \mathcal{N}(\mu_{i,t},\ \tau_i^{-1})$$

The models differ in the mean function $\mu_{i,t}$, the prior structure on α_i, and autoregressive component.

AR structure

An autoregressive component of lag 1 and 2 were initially added to the mean of the model. AR(1):

$$\begin{aligned} y_{i,t} &\sim \mathcal{N}\big(\mu_{i,t} + z_{i,t},\ \tau_i^{-1}\big), \quad t > 1 \\ \text{where} \quad z_{i,t} &= \phi\,(y_{i,t-1} - \mu_{i,t-1}) \\ \text{with} \quad \phi &\sim \text{Uniform}(-1, 1) \end{aligned}$$

AR(2):

$$\begin{aligned} &\phi_1(y_{i,t-1} - \mu_{i,t-1}) + \phi_2(y_{i,t-2} - \mu_{i,t-2}) \\ \text{with} \quad &\phi_1, \phi_2 \sim \text{Uniform}(-1, 1) \end{aligned}$$

Cycles

Added Fourier pairs to capture seasonal effects. A single harmonic with period P (weeks) is defined as:

$$\text{cycle}_i(t;\, P) = \beta_i \cos\big(\tfrac{2\pi t}{P}\big) + \gamma_i \sin\big(\tfrac{2\pi t}{P}\big)$$

Single-cycle models set $\text{cycles}_{i,t} = \text{cycle}_i(t;\, P)$ for $P \in \{52, 26, 8\}$ respectively.

Dual-cycle models sum two cycles with independent coefficients:

$$\text{cycles}_{i,t} = \text{cycle}_{1,i}(t;\, P_1) + \text{cycle}_{2,i}(t;\, P_2)$$

where $(P_1, P_2) \in \{(52, 26),\, (52, 8)\}$. The subscripts on $\text{cycle}_{1,i}$, $\text{cycle}_{2,i}$ correspond to independent coefficient pairs $(\beta_{1,i}, \gamma_{1,i})$ and $(\beta_{2,i}, \gamma_{2,i})$.

Event effects

Residual patterns motivated testing two event effects. A three week period around New Years (applied globally, or to each region) and a three-week Mid West funding reset (applied only to the Mid West region, $\text{mw}_i = \mathbf{1}(\text{region } i = \text{HSE Mid West})$). Each was fitted independently to confirm independent contribution:

Global New Year effect:

$$\text{ny}_t = \delta_{\text{pre}} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid}} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post}} \cdot \mathbf{1}(t \bmod 52 = 2)$$

Regional New Year effect:

$$\text{ny}_{i,t} = \delta_{\text{pre},i} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \cdot \mathbf{1}(t \bmod 52 = 2)$$

Mid West reset:

$$\text{mwr}_{i,t} = \sigma_{\text{pre}} \cdot \mathbf{1}(t = 86) \cdot \text{mw}_i + \sigma_{\text{mid}} \cdot \mathbf{1}(t = 87) \cdot \text{mw}_i + \sigma_{\text{post}} \cdot \mathbf{1}(t = 88) \cdot \text{mw}_i$$

Hierarchical intercepts

Replaces the independent priors on α_i with a hierarchical structure:

$$\alpha_i \sim \mathcal{N}(\mu_\alpha,\ \tau_\alpha^{-1})$$

with hyperpriors $\mu_\alpha \sim \mathcal{N}(0, 0.001)$ and $\tau_\alpha \sim \Gamma(0.001, 0.001)$. The full v5.1 mean function is given in Equation (7.1.1).

3.2 Response Scaling

The on trolley counts were standardised to 10,000 people in each health region, Raw trolley counts vary across regions primarily because of population size rather than differences in hospital pressure. To enable fair comparison, counts were standardised to a rate per 10,000 catchment population.

$$y_{i,t} = \frac{\text{weekly trolley count}_{i,t}}{\text{health region catchment population}_i} \times 10{,}000$$

This transforms the outcome into patients on trolley per 10,000 population per week, removing the confounding effect of region size and allowing direct comparison of hospital pressure across the six HSE health regions.

3.3 Posterior Sampling and Bayesian Ranking

Just another Gibbs sampler (*JAGs*), implemented using pyJAGS (Nowotny, 2024), a Python interface to JAGS (Plummer, 2003), was used to sample from the posterior distribution of our parameters given our data. Health region baselines, α_j^s, were used for ranking. Where j is one of the 6 HSE health regions, $j \in \{1, ..., 6\}$, and s is the draw index, for $s \in \{1, ..., n\}$ and total samples is defined as $n = 20{,}000$.

$$A = \begin{bmatrix} \alpha_1^1 & \cdots & \alpha_j^1 & \cdots & \alpha_6^1 \\ \vdots & & \vdots & & \\ \alpha_1^s & \cdots & \alpha_j^s & \cdots & \alpha_6^s \\ \vdots & & \vdots & & \\ \alpha_1^n & \cdots & \alpha_j^n & \cdots & \alpha_6^n \end{bmatrix}$$

Ranks follow the posterior ranking approach of Laird & Louis (1989), as applied by MacDermott et al. (2025). Where each sample is posterior sample is ranked to form rank distribution, R.

$$R = \begin{bmatrix} r_1^1 & \cdots & r_j^1 & \cdots & r_6^1 \\ \vdots & & \vdots & & \\ r_1^s & \cdots & r_j^s & \cdots & r_6^s \\ \vdots & & \vdots & & \\ r_1^n & \cdots & r_j^n & \cdots & r_6^n \end{bmatrix}$$

Where $r_j^s = \text{rank}(\alpha_j^s) = \sum_{i=1}^{6} I(\alpha_j^s > \alpha_i^s)$. Given a tie, ranks are increased by $\frac{1}{2}$ steps for each tied value:

$$r_j^s = \sum_{i=1}^{6} \left[\mathbf{1}(\alpha_j^s > \alpha_i^s) + \tfrac{1}{2}\mathbf{1}(\alpha_j^s = \alpha_i^s) \right]$$

Then posterior rank averages, $\hat{r}_i$, are calculated taking the mean across samples.

$$\hat{r}_j = \frac{1}{n} \sum_{s=1}^{n} r_j^s$$

$$\hat{R} = \left[\; \hat{r}_1, \hat{r}_2, \hat{r}_3, \hat{r}_4, \hat{r}_5, \hat{r}_6 \; \right]$$

Then posterior ranked averages are ranked once again to achieve ranked posterior ranked averages are calculated.

$$r_j^* = rank(\alpha_j^*) = \sum_{i=1}^{6} I(\alpha_j^* > \alpha_i^*)$$

$$R^* = \left[\; r_1^*, r_2^*, r_3^*, r_4^*, r_5^*, r_6^* \; \right]$$

Where $r_j^* = rank(\hat{r}_j) = \sum_{i=1}^{6} I(\hat{r}_j > \hat{r}_i)$.

4 Results

4.1 Model Comparison

Model fit was compared using the Deviance Information Criterion (DIC). A DIC table was generated at every response scale.

4.1.1 Scaled by 2022 health region population (per 10,000 people)

Table 1: DIC comparison across the models with a trolley rate per 10,000 people. Bold rows indicate the selected spine.

Model	Description	Deviance	pD	DIC
v4.1	**Regional NY + MW Reset**	**4555.9**	**93.6**	**4649.5**
v5.1	**Hierarchical Intercepts**	**4556.2**	**94.1**	**4650.3**
v3.2	**Regional New Year Effect**	**4604.6**	**87.3**	**4691.9**
v3.1	Global New Year Effect	4682.7	57.5	4740.2
v3.3	Mid West Reset Only	5259.2	57.2	5316.4
v2.1	**AR(1) + Annual Cycle**	**5294.9**	**51.2**	**5346.1**
v0.2	AR(2) Only	5331.5	28.6	5360.1
v2.5	Annual + 2-Month Cycle	5284.9	75.3	5360.2
v0.1	**AR(1) Only**	**5344.1**	**26.5**	**5370.6**
v2.4	Annual + 6-Month Cycle	5301.3	75.8	5377.1
v2.3	2 Month Cycle Only	5335.3	50.9	5386.1
v2.2	6 Month Cycle Only	5349.4	51.1	5400.5

4.1.2 Regional Baseline

4.1.3 Week-to-Week Autocorrelation

4.1.4 Annual Cycle

4.1.5 Regional Rankings

5 Discussion

6 References

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Nowotny, M. (2024). *pyJAGS: A Python interface to JAGS* (Version 2.4.0) [Computer software]. https://github.com/michaelnowotny/pyjags

Plummer, M. (2003). JAGS: A program for analysis of Bayesian graphical models using Gibbs sampling. *Proceedings of the 3rd International Workshop on Distributed Statistical Computing (DSC 2003)*. https://www.r-project.org/conferences/DSC-2003/Proceedings/Plummer.pdf

7 Appendix

Todo: full posterior trace plots, Gelman stats, residual diagnostics.

7.1 Model Inheritance

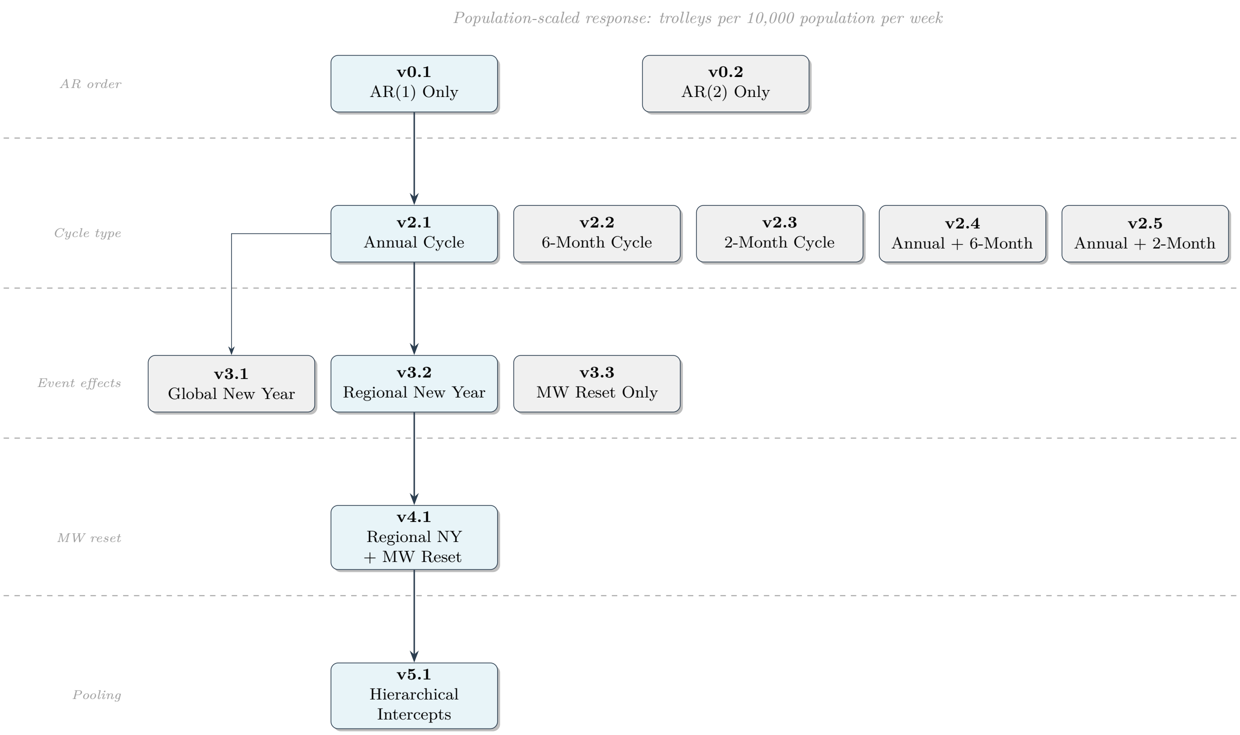

Figure 2: Model inheritance diagram. Spine nodes (blue) form the selected model sequence; grey nodes are tested variants. Thick arrows indicate the selected path; thin arrows indicate branches.

Full Model Specifications

Let $y_{i,t}$ denote the scaled trolley count for region i at week t, for $i \in \{1, \ldots, 6\}$ and $t \in \{1, \ldots, T\}$.

Model v0.1

AR(1) component with a baseline. Constant region-level intercept, no cycle, no event effects.

Likelihood

$$y_{i,1} \sim \mathcal{N}(\alpha_i,\ \tau_i^{-1})$$
$$y_{i,t} \sim \mathcal{N}\big(\alpha_i + \phi\,(y_{i,t-1} - \alpha_i),\ \tau_i^{-1}\big)\,, \quad t > 1$$

Mean function

$$\mu_{i,t} = \alpha_i$$

Priors

$$
\begin{aligned}
\alpha_i &\sim \mathcal{N}(0,\ 0.001) \\
\tau_i &\sim \Gamma(0.001,\ 0.001) \\
\phi &\sim \text{Uniform}(-1,\ 1)
\end{aligned}
$$

Model v0.2

AR(2) baseline. Extends v0.1 with a second autoregressive lag.

Likelihood

$$
\begin{aligned}
y_{i,1} &\sim \mathcal{N}(\alpha_i,\ \tau_i^{-1}) \\
y_{i,2} &\sim \mathcal{N}\big(\alpha_i + \phi_1\,(y_{i,1} - \alpha_i),\ \tau_i^{-1}\big) \\
y_{i,t} &\sim \mathcal{N}\big(\alpha_i + \phi_1\,(y_{i,t-1} - \alpha_i) + \phi_2\,(y_{i,t-2} - \alpha_i),\ \tau_i^{-1}\big)\,, \quad t > 2
\end{aligned}
$$

Mean function

$$
\mu_{i,t} = \alpha_i
$$

Priors

$$
\begin{aligned}
\alpha_i &\sim \mathcal{N}(0,\ 0.001) \\
\tau_i &\sim \Gamma(0.001,\ 0.001) \\
\phi_1,\ \phi_2 &\sim \text{Uniform}(-1,\ 1)
\end{aligned}
$$

Model v2.1

Extends v0.1. Adds a region-specific 52-week (annual) cycle.

Likelihood

$$
\begin{aligned}
y_{i,1} &\sim \mathcal{N}(\mu_{i,1},\ \tau_i^{-1}) \\
y_{i,t} &\sim \mathcal{N}\big(\mu_{i,t} + \phi\,(y_{i,t-1} - \mu_{i,t-1}),\ \tau_i^{-1}\big)\,, \quad t > 1
\end{aligned}
$$

Mean function

$$
\mu_{i,t} = \alpha_i + \beta_i \cos\!\big(\tfrac{2\pi t}{52}\big) + \gamma_i \sin\!\big(\tfrac{2\pi t}{52}\big)
$$

Priors

$$
\begin{aligned}
\alpha_i,\ \beta_i,\ \gamma_i &\sim \mathcal{N}(0,\ 0.001) \\
\tau_i &\sim \Gamma(0.001,\ 0.001) \\
\phi &\sim \text{Uniform}(-1,\ 1)
\end{aligned}
$$

Model v2.2

Extends v0.1. Adds a region-specific 26-week (semi-annual) cycle.

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\tfrac{2\pi t}{26}\right) + \gamma_i \sin\left(\tfrac{2\pi t}{26}\right)$$

Likelihood and priors identical to v2.1.

Model v2.3

Extends v0.1. Adds a region-specific 8-week cycle (pre-computed harmonic vectors).

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \ \cos_8[t] + \gamma_i \ \sin_8[t]$$

where $\cos_8[t] = \cos(2\pi t/8)$ and $\sin_8[t] = \sin(2\pi t/8)$.

Likelihood and priors identical to v2.1.

Model v2.4

Extends v2.1. Adds a second harmonic at 26 weeks (dual cycle: annual + semi-annual).

Mean function

$$\mu_{i,t} = \alpha_i + \beta_{1,i} \cos\left(\tfrac{2\pi t}{52}\right) + \gamma_{1,i} \sin\left(\tfrac{2\pi t}{52}\right) + \beta_{2,i} \cos_{26}[t] + \gamma_{2,i} \sin_{26}[t]$$

Priors

$$\begin{aligned} \alpha_i,\ \beta_{1,i},\ \gamma_{1,i},\ \beta_{2,i},\ \gamma_{2,i} &\sim \mathcal{N}(0,\ 0.001) \\ \tau_i &\sim \Gamma(0.001,\ 0.001) \\ \phi &\sim \text{Uniform}(-1,\ 1) \end{aligned}$$

Model v2.5

Extends v2.1. Adds a second harmonic at 8 weeks (dual cycle: annual + 8-week).

Mean function

$$\mu_{i,t} = \alpha_i + \beta_{1,i} \cos\left(\tfrac{2\pi t}{52}\right) + \gamma_{1,i} \sin\left(\tfrac{2\pi t}{52}\right) + \beta_{2,i} \cos_8[t] + \gamma_{2,i} \sin_8[t]$$

Priors identical to v2.4.

Model v3.1

Extends v2.1. Adds a global (shared across all regions) three-week New Year effect.

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\tfrac{2\pi t}{52}\right) + \gamma_i \sin\left(\tfrac{2\pi t}{52}\right) + \delta_{\text{pre}} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid}} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post}} \cdot \mathbf{1}(t \bmod 52 = 2)$$

Priors

$$\begin{aligned}\alpha_i,\ \beta_i,\ \gamma_i &\sim \mathcal{N}(0,\ 0.001)\\ \tau_i &\sim \Gamma(0.001,\ 0.001)\\ \phi &\sim \mathrm{Uniform}(-1,\ 1)\\ \delta_{\mathrm{pre}},\ \delta_{\mathrm{mid}},\ \delta_{\mathrm{post}} &\sim \mathcal{N}(0,\ 0.001)\end{aligned}$$

Model v3.2

Extends v3.1. New Year effect is region-specific rather than global.

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\tfrac{2\pi t}{52}\right) + \gamma_i \sin\left(\tfrac{2\pi t}{52}\right) + \delta_{\mathrm{pre},i}\cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\mathrm{mid},i}\cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\mathrm{post},i}\cdot \mathbf{1}(t \bmod 52 = 2)$$

Priors

$$\begin{aligned}\alpha_i,\ \beta_i,\ \gamma_i,\ \delta_{\mathrm{pre},i},\ \delta_{\mathrm{mid},i},\ \delta_{\mathrm{post},i} &\sim \mathcal{N}(0,\ 0.001)\\ \tau_i &\sim \Gamma(0.001,\ 0.001)\\ \phi &\sim \mathrm{Uniform}(-1,\ 1)\end{aligned}$$

Model v3.3

Extends v2.1. Adds a global Mid West funding reset effect (no New Year effect).

Mean function

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\tfrac{2\pi t}{52}\right) + \gamma_i \sin\left(\tfrac{2\pi t}{52}\right) + \sigma_{\mathrm{pre}}\cdot \mathbf{1}(t = 86)\cdot \mathrm{mw}_i + \sigma_{\mathrm{mid}}\cdot \mathbf{1}(t = 87)\cdot \mathrm{mw}_i + \sigma_{\mathrm{post}}\cdot \mathbf{1}(t = 88)\cdot \mathrm{mw}_i$$

where $\mathrm{mw}_i = \mathbf{1}(\text{region } i = \text{HSE Mid West})$.

Priors

$$\begin{aligned}\alpha_i,\ \beta_i,\ \gamma_i &\sim \mathcal{N}(0,\ 0.001)\\ \tau_i &\sim \Gamma(0.001,\ 0.001)\\ \phi &\sim \mathrm{Uniform}(-1,\ 1)\\ \sigma_{\mathrm{pre}},\ \sigma_{\mathrm{mid}},\ \sigma_{\mathrm{post}} &\sim \mathcal{N}(0,\ 0.001)\end{aligned}$$

Model v4.1

Extends v3.2. Combines region-level New Year effects with the global Mid West funding reset.

Mean function

$$\begin{aligned}\mu_{i,t} = \alpha_i &+ \beta_i \cos\left(\tfrac{2\pi t}{52}\right) + \gamma_i \sin\left(\tfrac{2\pi t}{52}\right)\\ &+ \delta_{\mathrm{pre},i}\cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\mathrm{mid},i}\cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\mathrm{post},i}\cdot \mathbf{1}(t \bmod 52 = 2)\\ &+ \sigma_{\mathrm{pre}}\cdot \mathbf{1}(t = 86)\cdot \mathrm{mw}_i + \sigma_{\mathrm{mid}}\cdot \mathbf{1}(t = 87)\cdot \mathrm{mw}_i + \sigma_{\mathrm{post}}\cdot \mathbf{1}(t = 88)\cdot \mathrm{mw}_i\end{aligned}$$

Priors

$$\alpha_i,\ \beta_i,\ \gamma_i,\ \delta_{\text{pre},i},\ \delta_{\text{mid},i},\ \delta_{\text{post},i} \sim \mathcal{N}(0,\ 0.001)$$
$$\tau_i \sim \Gamma(0.001,\ 0.001)$$
$$\phi \sim \text{Uniform}(-1,\ 1)$$
$$\sigma_{\text{pre}},\ \sigma_{\text{mid}},\ \sigma_{\text{post}} \sim \mathcal{N}(0,\ 0.001)$$

7.1.1 Model v5.1

Let $y_{i,t}$ denote the scaled trolley count for region i at week t, for $i \in \{1, \dots, 6\}$ and $t \in \{1, \dots, T\}$.

Likelihood

$$y_{i,1} \sim \mathcal{N}(\mu_{i,1},\ \tau_i^{-1})$$
$$y_{i,t} \sim \mathcal{N}\big(\mu_{i,t} + \phi\,(y_{i,t-1} - \mu_{i,t-1}),\ \tau_i^{-1}\big)\,, \quad t > 1$$

Mean function

$$\begin{aligned}\mu_{i,t} = {} & \alpha_i + \beta_i \cos\!\big(\tfrac{2\pi t}{52}\big) + \gamma_i \sin\!\big(\tfrac{2\pi t}{52}\big) \\ & + \delta_{\text{pre},i} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \cdot \mathbf{1}(t \bmod 52 = 2) \\ & + \sigma_{\text{pre}} \cdot \mathbf{1}(t = 86) \cdot \text{mw}_i + \sigma_{\text{mid}} \cdot \mathbf{1}(t = 87) \cdot \text{mw}_i + \sigma_{\text{post}} \cdot \mathbf{1}(t = 88) \cdot \text{mw}_i\end{aligned}$$

Model v5.1 mean function

Where $\text{mw}_i = \mathbf{1}(\text{region } i = \text{HSE Mid West})$. The β_i and γ_i coefficients capture the annual seasonal cycle, $\frac{\gamma_i}{\beta_i}$. The $\delta_{\text{pre/mid/post},i}$ components are region level new year effects. The $\sigma_\cdot$ component captures the Mid West funding reset.

The term $\phi\,(y_{i,t-1} - \mu_{i,t-1})$ in Equation (7.1.1) adds an AR(1) component, allowing each weeks observation to depend on the previous week's derivation from the mean.

Priors

The baseline parameters α_i are given a hierarchical prior, treating the six HSE regions as exchangeable within the same national health system:

$$\alpha_i \sim \mathcal{N}(\mu_\alpha,\ \tau_\alpha^{-1})$$
$$\mu_\alpha \sim \mathcal{N}(0,\ 0.001)$$
$$\tau_\alpha \sim \Gamma(0.001,\ 0.001)$$

Remaining priors are independent and uninformative:

$$\beta_i,\ \gamma_i,\ \delta_{\text{pre},i},\ \delta_{\text{mid},i},\ \delta_{\text{post},i} \sim \mathcal{N}(0,\ 0.001)$$
$$\tau_i \sim \Gamma(0.001,\ 0.001)$$
$$\phi \sim \text{Uniform}(-1,\ 1)$$
$$\sigma_{\text{pre}},\ \sigma_{\text{mid}},\ \sigma_{\text{post}} \sim \mathcal{N}(0,\ 0.001)$$

Table 2: DIC comparison across the models (trolley count scaled by 2026 regional budget). Bold rows indicate the selected spine.

Model	Description	Deviance	pD	DIC
v5.1	**Hierarchical Intercepts**	**19858.3**	**80.6**	**19938.9**
v3.1	Global New Year Effect	19920.3	54.9	19975.3
v4.1	**Regional NY + MW Reset**	**19899.9**	**79.4**	**19979.3**
v3.2	**Regional New Year Effect**	**19908.3**	**77.4**	**19985.7**
v3.3	Mid West Reset Only	20169.5	50.4	20219.9
v2.1	**AR(1) + Annual Cycle**	**20176.0**	**48.6**	**20224.6**
v2.5	Annual + 2-Month Cycle	20171.4	72.1	20243.5
v2.4	Annual + 6-Month Cycle	20195.4	70.6	20266.0
v0.1	**AR(1) Only**	**20253.9**	**27.0**	**20280.9**
v0.2	AR(2) Only	20253.9	29.3	20283.2
v2.3	2 Month Cycle Only	20248.9	50.6	20299.5
v2.2	6 Month Cycle Only	20274.2	48.8	20323.0

7.2 Alternative Response Scales

The primary analysis uses population-scaled trolley counts (per 10,000 catchment population). Two sensitivity scalings were also fitted to assess robustness of model selection and regional rankings to the choice of denominator.

7.2.1 Budget scaling

Weekly trolley counts scaled by the 2026 HSE regional budget allocation, expressing the outcome as trolleys per €1 billion allocated to the region.

$$y_{i,t} = \frac{\text{weekly trolley count}_{i,t}}{\text{health region budget}_i / 10^6}$$

where health region budget$_i$ is the 2026 budget allocation for region i in €thousands. This controls for funding differentials: a region receiving a larger allocation is expected to handle a proportionally larger patient volume.

7.2.2 Bed scaling

Weekly trolley counts scaled by the number of inpatient beds in each region, expressing the outcome as an overcapacity rate: for every 100 inpatient beds, how many patients were waiting on trolleys that week.

$$y_{i,t} = \frac{\text{weekly trolley count}_{i,t}}{\text{health region inpatient beds}_i} \times 100$$

where health region inpatient beds$_i$ is the total inpatient bed count across hospitals in region i, summed from the June 2025 bed census. Hospitals with missing bed counts are excluded from the regional sum.

Table 3: DIC comparison across models (response scaled by June 2025 health region inpatient bed count). Bold rows indicate the selected spine.

Model	Description	Deviance	pD	DIC
v4.1	**Regional NY + MW Reset**	**11448.9**	**93.0**	**11541.9**
v5.1	**Hierarchical Intercepts**	**11449.0**	**93.3**	**11542.4**
v3.2	**Regional New Year Effect**	**11470.0**	**87.3**	**11557.3**
v3.1	Global New Year Effect	11510.1	57.2	11567.3
v3.3	Mid West Reset Only	11738.6	56.7	11795.3
v2.1	**AR(1) + Annual Cycle**	**11756.7**	**51.0**	**11807.7**
v2.5	Annual + 2-Month Cycle	11751.8	75.1	11826.9
v2.4	Annual + 6-Month Cycle	11773.3	75.3	11848.6
v0.1	**AR(1) Only**	**11828.7**	**26.6**	**11855.3**
v0.2	AR(2) Only	11828.3	28.6	11856.8
v2.3	2 Month Cycle Only	11823.4	50.7	11874.1
v2.2	6 Month Cycle Only	11845.5	50.9	11896.4